

- 2 (a) Fig. 2.1 is an incomplete diagram of the structure of an α -glucose molecule.
- Complete Fig. 2.1 to show the structure of an α -glucose molecule.

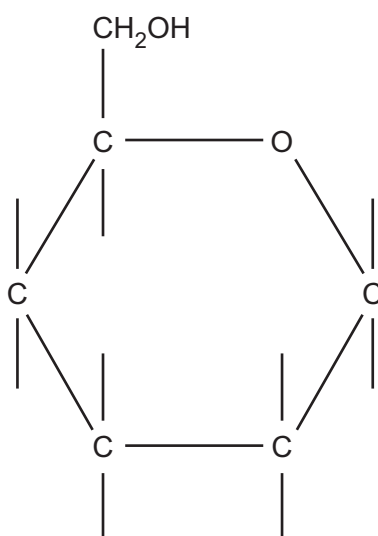


Fig. 2.1

[2]

- (b) Fig. 2.2 shows part of a glycoprotein molecule.

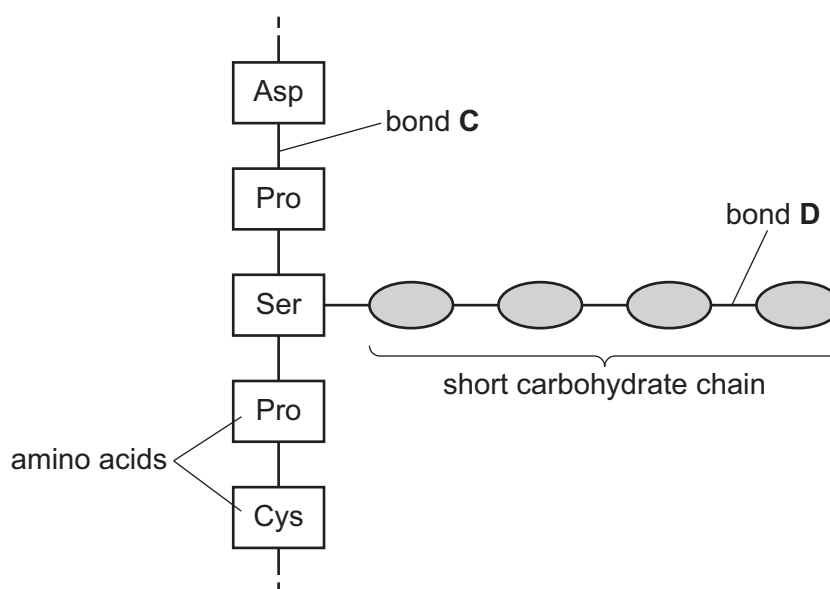


Fig. 2.2

- (i) State the types of covalent bond labelled **C** and **D** in Fig. 2.2.

C

D

[2]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

(ii) Many glycoproteins in the cell surface membrane are involved in cell signalling.

State the role in cell signalling of glycoproteins in cell surface membranes.

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(c) One of the proteins found in milk is β -casein.

A molecule of β -casein consists of a single polypeptide of approximately 200 amino acids.

Molecules of β -casein have a **high** proportion of the amino acids proline and leucine. These amino acids have hydrophobic R-groups.

Fig. 2.3 compares the structure of proline (Pro) with the general structure of an amino acid. This shows that proline is an unusual amino acid because it has a cyclic R-group and the nitrogen atom is attached to only one hydrogen atom.

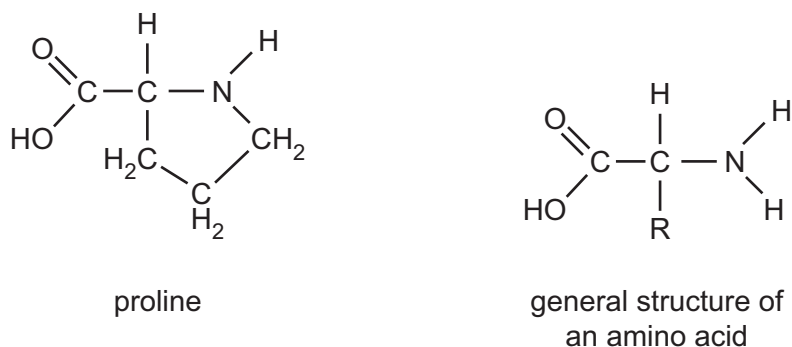


Fig. 2.3

When a molecule of proline becomes part of a polypeptide, the hydrogen atom is lost from the nitrogen atom and is therefore **not** available for the formation of hydrogen bonds.

Molecules of β -casein have a relatively **low** proportion of the amino acids cysteine and serine. Table 2.1 shows the R-groups of cysteine and serine.

Table 2.1

amino acid	R-group	feature of R-group
cysteine (Cys)	$\text{—CH}_2\text{SH}$	can form a disulfide bond
serine (Ser)	$\text{—CH}_2\text{OH}$	hydroxyl group present



Compared to other protein molecules with a similar number of amino acids, β -casein molecules have:

- a much less organised secondary structure
- relatively little tertiary structure.

With reference to the four named amino acids, proline, leucine, serine and cysteine, suggest possible explanations for these **two** observations.

less organised secondary structure

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relatively little tertiary structure

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[4]

[Total: 9]